

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination

May 2012

CL101.01 Fundamentals of Civil Engineering

Date: 11.05.2012, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Enlist basic requirements of a building planning. [02]
(b) Explain functions of civil engineering. [05]
- Q - 2 (a) Explain light plane and set-back with neat sketch. [04]
(b) Give classification of building based on occupancy. [04]
(c) Write advantages and dis-advantages of railways. [06]

OR

- Q - 2 (a) Briefly describe different traffic sign. [04]
(b) Explain Nagpur Road Plan in detail. [02]
(c) Write a short note on Roominess and Circulation. [04]
(d) Give the functions of Roof, Column, Window, and Plinth. [04]
- Q - 3 (a) Define Superstructure and Foundation soil. [02]
(b) Write a note on Wind load and Earthquake load. [04]
(c) Draw a neat sketch of 30 cm thick wall foundation showing all dimensions. [02]
(d) Write the requirements and uses of stones. [06]

OR

- Q - 3 (a) Draw the detailed section at A-A for the plan shown in **Figure 1**. Take suitable scale. [08]
(b) Enlist properties of a good timber. [02]
(c) What is concrete? Write advantages and dis-advantages of concrete. [04]

SECTION - II

- Q - 4 (a) Briefly describe the sources of water. [03]
(b) While tracing the perimeter of a rectangle of size 15 cm x 10 cm with the anchor point inside the rectangle and tracing arm set to the natural scale of planimeter, the initial reading is 8.764. The zero of the counting disc passed the index mark twice in the anticlockwise direction. Find the area of zero circle. [04]

- Q - 5** During a closed compass traverse survey, the observed F.B. and B.B. are noted in table below. Determine which stations are affected by local attraction. Find out included angles and corrected bearings. Also apply necessary checks. [14]

Line	F.B.	B.B.
AB	74° 30'	256° 00'
BC	107° 30'	286° 30'
CD	224° 30'	44° 30'
DA	306° 30'	126° 00'

OR

- Q - 5 (a)** Explain the following term: [04]

(i) Dip of Needle (ii) Declination (iii) Isogonic lines (iv) True bearing.

- (b)** Explain principles of surveying. [04]

- (c)** Discuss the methods of ranging in detail. [06]

- Q - 6 (a)** The following consecutive readings were taken with a level and levelling staff on continuously sloping ground at an interval of 20 m. [07]

0.385 1.030 1.925 2.825 3.730 4.685 0.625 2.005 3.110 4.485

The R.L. of first point is 208.125 m. Calculate the reduced levels by using Rise and Fall method. Prepare a page of level book.

- (b)** Discuss water conveyance system in detail. [07]

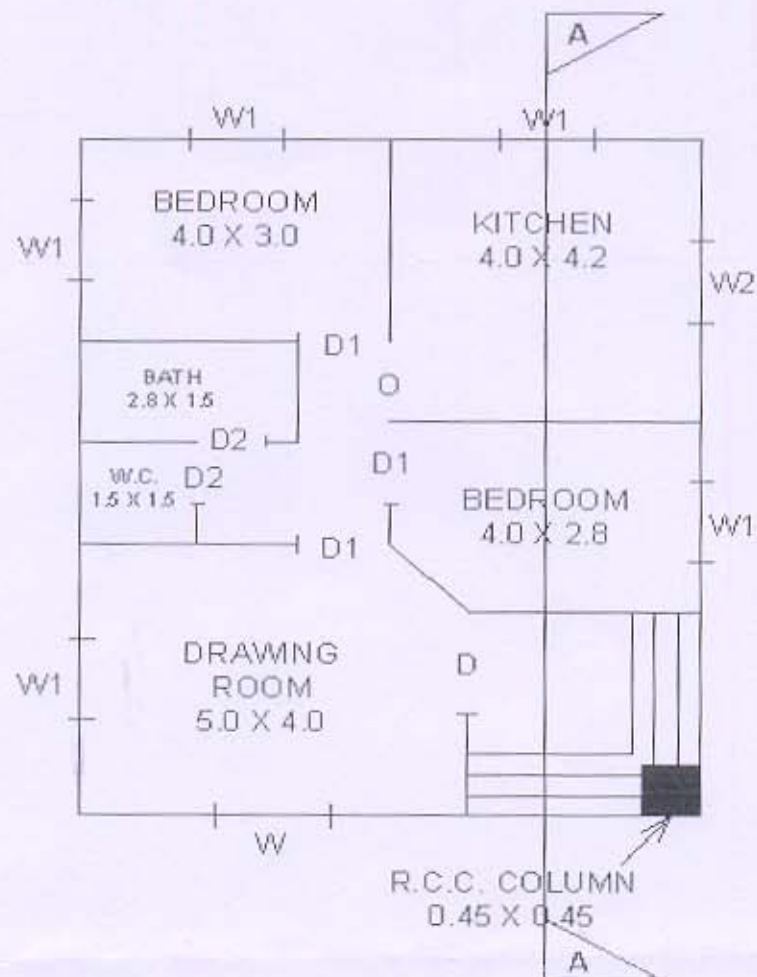
OR

- Q - 6 (a)** Calculate the volume of earthwork in an embankment for which the cross sectional areas at 20 m interval are as shown in table below. [04]

Distance (m)	Area (m ²)
0	38
20	62
40	74
60	18
80	22
100	28
120	13

- (b)** Discuss about water requirements. Explain how conservation of water is carried out? [07]

- (c)** Write steps for adjustment of a dumpy level. [03]



SCHEDULE OF OPENING		
SR.NO.	DESCRIPTION	SEE
1	D	1.5 X 2.1
2	D1	1.2 X 2.1
3	D2	0.9 X 2.1
4	O	1.2 X 2.1
5	W	1.5 X 1.2
6	W1	1.2 X 1.2
7	W2	1.2 X 0.9

RISER = 0.15 m

TRADE = 0.30 m

FLOOR HEIGHT = 3.0 m

All dimensions are in meter (m).

Figure 1